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# UNIVERSITY OF CAPE TOWN

Test - 17 March 2018

Portfolio Optimization: ECO 4122Z

2  $\frac{1}{2}$  Hours

Maximum Marks: 75

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## INSTRUCTIONS TO CANDIDATES

Answer ALL questions

1. During the past five years, you owned two stocks that had the following annual rates of return:

| Year | Stock A | Stock B |
|------|---------|---------|
| 1    | 0.19    | 0.08    |
| 2    | 0.08    | 0.03    |
| 3    | -0.12   | -0.09   |
| 4    | 0.03    | 0.02    |
| 5    | 0.15    | 0.04    |

- (a) Compute the arithmetic mean annual rate of return for each stock. Which stock is most desirable by this measure? [4 marks]
- (b) Compute the standard deviation of the annual rate of return for each stock. By this measure, which is the preferable stock? [5 marks]
- (c) Compute the coefficient of variation for each stock. By this relative measure of risk, which stock is preferable? [3 marks]
- (d) Compute the geometric mean rate of return for each stock. Discuss the difference between the arithmetic mean return and the geometric mean return for each stock. Discuss the differences in the mean returns relative to the standard deviation of the return for each stock. [6 marks]

[Total: 18 marks]

2. (a) State the main assumptions of mean-variance portfolio theory. [4 marks]

Consider a mean-variance portfolio model with two securities, with respective returns  $R_A$  and  $R_B$ , where the expected return  $E[R_B] = 0.25E[R_A]$  and the variance of return  $Var(R_B) = 0.25Var(R_A)$ . Let the correlation between the returns on the two securities be  $\rho$ .

- (b) Determine, in terms of  $E[R_A]$ , the expected return on the minimum variance portfolio if:

- i.  $\rho = 0$
- ii.  $\rho = 1$

[2;2 marks]

- (c) i. Calculate the variance of the return on the minimum variance portfolio for part (b)(ii).  
 ii. Comment on the risk in this portfolio.

[2;2 marks]

[Total: 10 marks]

3. Suppose you derive the following information for the companies listed below, based on five years of monthly data:

| Company                | $\alpha_i$ ( Intercept) | $\sigma_i$ | $r_{iM}$ |
|------------------------|-------------------------|------------|----------|
| ABSA                   | 0.22                    | 12.10%     | 0.72     |
| Anglo America          | 0.10                    | 14.60      | 0.33     |
| MTN                    | 0.17                    | 7.60       | 0.55     |
| Shoprite               | 0.05                    | 10.20      | 0.60     |
| All Share Index (ALSI) | 0.00                    | 5.50       | 1.00     |

where  $\sigma_i$  is standard deviation for asset  $i$  and  $r_{iM}$  is the correlation between asset  $i$  and the market ( $M$ ).

- (a) Compute the beta coefficient for each stock. Comment on whether each stock is aggressive or defensive.

[4 marks]

- (b) Assuming a risk-free rate of 8 percent and an expected return for the market portfolio of 15 percent, compute the expected (required) return for all the stocks and plot them on the SML.

[6 marks]

- (c) Plot the following estimated returns for the next year on the SML and indicate which stocks are undervalued or overvalued, assuming the betas remain unchanged.

- ABSA - 20 percent
- Anglo American -15 percent
- MTN -19 percent
- Shoprite -10 percent

[5 marks]

[Total 15 marks]

4. You are the lead manager of a large mutual fund. You have become aware that several equity analysts who have recently joined your management team are interested in understanding the differences between the capital asset pricing model (CAPM) and arbitrage pricing theory (APT). In particular, they are interested in how these two asset pricing models can help them perform better security analysis.

- (a) Explain what the CAPM and APT attempt to model. What is the main difference between these two asset pricing models? [4;2 marks]
- (b) Under what circumstances would the APT be preferred over the CAPM as a tool for selecting stocks for the fund portfolio? [3 marks]

[Total: 9 marks]

5. An investor measures the utility of her wealth( $w$ ) using the utility function  $U(w) = \ln(w)$  for  $w > 0$ .

- (a) Derive the absolute and relative risk aversions for this investor's utility function, and the first derivative of each. [3 marks]
- (b) Comment on what this tells us about the proportion of her assets that this investor will invest in risky assets. [2 marks]

[Total: 5 marks]

6. Let  $R_i$  denote the return on security  $i$  given by the following multifactor model:

$$R_i = \alpha_i + b_{i,1}I_1 + b_{i,2}I_2 \cdots b_{i,L}I_L + \epsilon_i$$

where  $\alpha_i$  and  $\epsilon_i$  are the constant and random parts respectively of the component of the return unique to security  $i$ ,  $I_1, \dots, I_L$  are the changes in a set of the  $L$  indices and  $b_{i,k}$  is the sensitivity (factor beta) of security  $i$  to factor  $k$ .

- (a) State the category of the above model where:
- i. index 1 is a price index, index 2 is the yield on government bonds and index 3 is the annual rate of economic growth.
  - ii. index 1 is the level of Research and Development expenditure, index 2 is the price earnings ratio, index 3 is the level of gearing. [1;1 marks]

Consider the following two-factor model for the returns on three assets  $A, B$  and  $C$ :

| Asset      | A    | B    | C   |
|------------|------|------|-----|
| $\alpha_i$ | 0.03 | 0.05 | 0.1 |
| $b_{i,1}$  | 1    | 3    | 1.5 |
| $b_{i,2}$  | -4   | 2    | 1.5 |

(b) Determine the equation for the expected return on a portfolio which:

i. equally weights the three securities.

ii. has weights  $x_A = -0.5, x_B = 1.5, x_C = 0$ . [4 marks]

(c) Construct a portfolio of securities  $A, B, C$  that has a factor beta of 2 on the first factor and 1 on the second factor, i.e. the expected return on the portfolio is:

$$R_P = \alpha_P + 2I_1 + I_2 + \epsilon_i$$

[3 marks]

[ Total: 9 marks]

7. (a) Define the three forms of the Efficient Markets Hypothesis (EMH). [3 marks]

(b) State two reasons why it is hard to test whether any of the three forms hold in practice. [2 marks]

(c) Discuss the following two scenarios in the light of the EMH:

i. **Scenario 1:** The share price of Komatsu Limited, a company listed in the Tokyo Stock Exchange, Japan, fall suddenly, immediately after news of a Tsumani in the capital city of one of its major markets.

ii. **Scenario 2:** The share price of Steinhoff International Holdings, a company listed in the JSE, falls suddenly, when information regarding accounting irregularities is made public. [2;2 marks]

[Total: 9 marks]